Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/01/2004
PI Name:	Barcellos-Hoff, Mary Helen Ph.D.		
Project Title:	NSCOR: Mechanisms of HZE Damage and Repair in Human Epithelial Cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
Start Date:	12/01/2003	**End Date:**	11/30/2008
No. of Post Docs:	5	**No. of PhD Degrees:**	
No. of PhD Candidates:	12	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	NNA04CF75I		
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Performance Goal Text:			

Task Description:	NASA's Biomedical Critical Path Roadmap defines the carcinogenic risks of radiation exposure as one of only four Type I risks identified. Estimates of cancer risk from human travel in space should be based on a mechanistic understanding of complex effects that different types of radiation exposure elicit. Because most adult human solid tumors are epithelial in origin and because radiation effects are dictated by cellular genotype and phenotype, studies of HZE particle radiation biology using a physiologically relevant model will be the most informative regarding risk. We have developed methods for culturing human mammary epithelial cells (HMEC) derived from normal tissue as both monolayer and multicellular aggregates, which enable study of phenotypes and pathways that may be uniquely human, epithelial or multicellular. We propose to study how controlled cellular and microenvironment conditions affect early responses to HZE particle radiation-induced DNA damage in terms of repair, foci formation and gene expression, and persistent radiation-induced phenotypes and pathways that precede neoplasia. The potential for inducing genomic instability, which may contribute to its carcinogenic effect, will be measured in both finite lifespan HMEC and immortal HMEC. Together, these surrogate endpoints will enable functional evaluation of neoplastic potential as a function of radiation quality, energy and dose. These data will be integrated at two levels: by theoretical modeling of the physical events leading to DNA damage and by systems biology modeling of critical pathways. The intent of this 5-year project is to provide a comprehensive examination of HZE effects from the initial damage, to early cellular HMEC responses, to persistent functional precursors of carcinogenesis. We will address the mechanisms, nature, and frequency of DNA damaging events; mechanisms of DNA repair and misrepair; early signal transduction mechanisms; immediate and long-term, reversible and irreversible gene expression changes; cellular remodeling and reorganization; potential mechanisms of tissue "repair" and matrix effects; and the cellular and molecular mechanisms of HZE/proton-induced progression to a neoplastic phenotype. To provide a comprehensive picture of HZE effects from the initial damage, to early cellular HMEC responses, to persistent functional precursors of carcinogenesis. Several NASA areas of interest will be addressed: mechanisms, nature, and frequency of DNA damaging events; mechanisms of DNA repair and misrepair; early signal transduction mechanisms; immediate and long-term, and reversible and irreversible gene expression changes; cellular remodeling and reorganization; potential mechanisms of tissue "repair" and matrix effects; induction and regulation of genomic instability; and the cellular and molecular mechanisms of charged particle-induced progression to a neoplastic phenotype. The data obtained in these studies will be integrated at two levels: by theoretical modeling of the physical events leading to DNA damage and by systems biology modeling of critical pathways. Ultimately, this work will help NASA develop a basic understanding of the mechanisms of HZE radiation damage and repair and their contributions to the neoplastic process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Although cancer is considered the major late health risk following low fluence heavy charged particle radiation exposure, it has a latency of years that is both difficult to track and poorly understood. The use of cultured human cells permits study of radiation-induced phenotypes and pathways that precede neoplasia. Preliminary data from our research demonstrate that sparsely ionizing radiation can elicit cell behaviors, such as loss of multicellular organization, that are strongly associated with epithelial cancer and deregulation of differentiation. Importantly, the irradiated phenotype is expressed in the progeny of irradiated HMEC. The use of well characterized human epithelial cells and functional assays of neoplastic behavior make the approach relevant to human health risk estimates. Participants in this NSCOR have developed and implemented unique cell culture methodologies that more accurately reflect the behavior of human epithelial cells in vivo. In the first year of the program, we have recruited 4 new staff members who have been trained to execute and analyze HZE radiation response using the HMEC models at NSRL 3 and 4. Cell culture and gene expression core resources were established. The data from the first two NSRL runs have determined the relative biological effectiveness for 90% clonal death. In addition, we have developed techniques and tools for imaging nuclear foci localized by immunofluorescence as a measure of DNA damage and repair. Theoretical modeling of DNA damage indicate that there are many more complex DNA aberrations from HZE radiation exposure than what is measured by experiments. These results suggest that underestimation of this damage may be a significant factor in risk estimations. The objective for the coming year is to use system biology to build an integrative model of data on the rapid and persistent responses to HZE radiation damage.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Barcellos-Hoff MH. "How tissues respond to damage at the cellular level." British Journal of Radiology. In press, as of January 2005. , Jan-2005
Presentation	Barcellos-Hoff, M.H. "Interaction of cell and tissue stress responses." 3rd International Workshop on Space Radiation Research, Port Jefferson, NY. May-2004
Presentation	Barcellos-Hoff, M.H. "Non-mutagenic actions of radiation in mouse and human mammary carcinogenesis." Keystone Symposia on Mouse Models of Cancer, Keystone, CO. Feb-2004
Presentation	Barcellos-Hoff, M.H. "Basic studies of solid tumor cancer risk assessment." Workshop on Solid Cancer Risks from Space Radiation, Houston, TX. Jan-2004
Presentation	Barcellos-Hoff, M.H. "Regulation of cell fate decision by TGF-ß following DNA damage." Graduate Program in Cell and Molecular Biology, Colorado State University, Ft. Collins, CO. Feb-2004